Logarithm

30 Hard-Level Questions from the Chapter (Solved Examples + Exercise) · 8 Extra Questions · Fully Worked Solutions

All 13 Solved Examples and all 92 Exercise questions of the chapter were checked one by one. The 30 questions below were chosen because they are the hardest ones and because, between them, they cover every pattern that keeps appearing in Bangladesh and Indian competitive exams (BCS, Bank, MBA/IBA admission, university admission, CAT, MAT, IIFT, CDS, RRB, SSC and similar). They are grouped by pattern and go from easier to harder inside each group. The **8 green cards (Q.31–Q.38)** are extra questions that are not in the PDF. They fill the gaps the PDF leaves: a variable base, inequalities, comparing logs, zeros after the decimal point, and proof-type identities.

Notation. When no base is written, $\log x$ means $\log_{10} x$, as in the PDF. Every solution is written step by step for a first-time learner. A **Key idea** box gives the reusable trick, and a **Trap** box gives the mistake examiners hope you will make.

Before We Begin: The Toolkit

1. What a logarithm really is

A logarithm is simply an **exponent**. The statement $\log_a x = m$ answers the question "to what power must I raise a to get x?"

$$a^m = x \quad \Longleftrightarrow \quad m = \log_a x$$

For example, $\log_2 8 = 3$ because $2^3 = 8$, and $\log_{10} 0.01 = -2$ because $10^{-2} = \frac{1}{100}$. Whenever a log problem confuses you, switch it back to this exponent form.

2. When does a logarithm exist?

$\log_a x$ is defined only when **all three** conditions hold:

$$a > 0, \qquad a \neq 1, \qquad x > 0.$$

So the log of zero or of a negative number does not exist, and the base can never be negative, zero or 1. Many "hard" questions are really testing these conditions (see Q.2, Q.23 and Q.33).

3. The laws you will use again and again

Name	Law (for valid a, b, x, y)
Product	$\log_a(xy) = \log_a x + \log_a y$
Quotient	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
Power	$\log_a x^p = p \log_a x$
Special values	$\log_a a = 1, \quad \log_a 1 = 0$
Change of base	$\log_a x = \frac{\log_b x}{\log_b a} = \frac{\log x}{\log a}$
Reciprocal	$\log_a x = \frac{1}{\log_x a}$
Chain	$\log_a b \cdot \log_b c = \log_a c$
Power in base	$\log_{a^q} x^p = \frac{p}{q} \log_a x$
Inverse	$a^{\log_a x} = x$
Swap	$x^{\log_a y} = y^{\log_a x}$

Almost every law comes from the change-of-base law. For instance, the chain law is just $\frac{\log b}{\log a} \times \frac{\log c}{\log b} = \frac{\log c}{\log a}$. If you forget a law, write every log as $\frac{\log(\text{number})}{\log(\text{base})}$ and simplify.

4. The graph: why equations and inequalities behave as they do

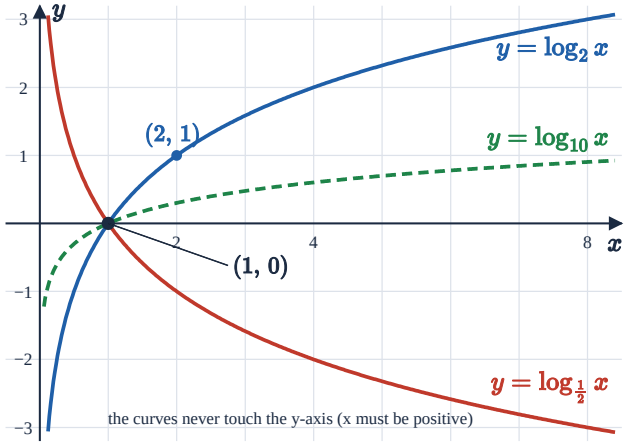


Figure 1. Graphs of $y = \log_a x$ for $a = 2$, $a = 10$ and $a = \frac{1}{2}$.

- Every curve passes through $(1, 0)$, because $\log_a 1 = 0$ for any base.
- If $a > 1$ the curve **rises**, so a bigger x gives a bigger log. If $0 < a < 1$ the curve **falls**.
- Each curve never meets the same height twice, so $\log_a A = \log_a B \Rightarrow A = B$. This is how log equations are solved.
- Because a curve with base below 1 falls, an inequality **reverses** when you remove a log whose base lies between 0 and 1 (Q.34).

5. Characteristic, mantissa and number of digits

Any common logarithm is written as an integer plus a decimal part lying in $[0, 1)$:

$$\log N = \underbrace{\text{characteristic}}_{\text{integer}} + \underbrace{\text{mantissa}}_{0 \le \cdot < 1}$$

- If $N \geq 1$ and has n digits before the decimal point, the characteristic is $n - 1$. So **the number of digits of N is its characteristic plus 1**.
- If $0 < N < 1$ and there are k zeros right after the decimal point, the characteristic is $-(k + 1)$. It is written with a bar, as $\overline{k+1}$; for example, $\log 0.0035$ has characteristic $\overline{3}$.
- The mantissa is **never negative**, and it depends only on the digits of N, not on where the decimal point is.

Values you are expected to know: $\log 2 = 0.30103$, $\log 3 = 0.47712$, $\log 7 = 0.84510$, and $\log 5 = 1 - \log 2 = 0.69897$.

Pattern	Questions
Surd and negative bases; relation between unknowns	1 – 3
Nested (log of a log) expressions	4, 5
Simplifying with the laws (factorising, telescoping)	6, 7
Conditions such as $\log x + \log y = \log(x+y)$	8, 9
The identity $a^{\log_a x} = x$	10, 32
Reciprocal sums equal to a constant	11 – 13, 37
Chain and cyclic products	14 – 17
One log in terms of given logs	18 – 21
Log equations (domain, mixed bases, surds, AP)	22 – 26, 31, 33
Characteristic, mantissa, bar notation	27, 28, 36
Number of digits	29, 30
Inequalities and comparison	34, 35
Proof-type identities	37, 38

Part A · Evaluating Logs with Awkward Bases

Q.1 Exercise Q.22 *Surd base*

The value of $\log_{2\sqrt{3}}(1728)$ is

(a) 3 (b) 5
(c) 6 (d) 9

SOLUTION

Step 1. Get rid of the root in the base. Squaring a surd base usually turns it into a whole number:

$$(2\sqrt{3})^2 = 2^2 \times (\sqrt{3})^2 = 4 \times 3 = 12.$$

Step 2. Write 1728 using 12. Since $12 \times 12 = 144$ and $144 \times 12 = 1728$, we have $1728 = 12^3$. Now replace 12 by $(2\sqrt{3})^2$:

$$1728 = 12^3 = \left[(2\sqrt{3})^2\right]^3 = (2\sqrt{3})^6.$$

Step 3. Apply the definition. We have written 1728 as an exact power of the base, so

$$\begin{aligned}\log_{2\sqrt{3}} 1728 &= \log_{2\sqrt{3}}(2\sqrt{3})^6 \\ &= 6\log_{2\sqrt{3}}(2\sqrt{3}) = 6 \times 1 = 6.\end{aligned}$$

Second method (change of base).

$$\log_{2\sqrt{3}} 1728 = \frac{\log 12^3}{\log 12^{\frac{1}{2}}} = \frac{3\log 12}{\frac{1}{2}\log 12} = 6.$$

Answer: (c) 6

KEY IDEA. Always try to write the number as a power of the base. When the base is a surd, square it first. The same trick gives Exercise Q.5: since $(3\sqrt{2})^2 = 18$, we get $\log_{3\sqrt{2}} \frac{1}{18} = \log_{3\sqrt{2}}(3\sqrt{2})^{-2} = -2$.

Q.2 Exercise Q.21 *Negative base (concept trap)*

The value of $\log_{(-\frac{1}{3})} 81$ is equal to

(a) -27 (b) -4
(c) 4 (d) 27

SOLUTION (THE EXAMINER'S METHOD)

Let the value be x. By the definition,

$$\left(-\frac{1}{3}\right)^x = 81 = 3^4.$$

Because 4 is even, $3^4 = (-3)^4$. Also -3 is the reciprocal of $-\frac{1}{3}$, so $-3 = \left(-\frac{1}{3}\right)^{-1}$. Therefore

$$81 = (-3)^4 = \left[\left(-\frac{1}{3}\right)^{-1}\right]^4 = \left(-\frac{1}{3}\right)^{-4}.$$

Comparing the powers gives $x = -4$.

SOLUTION (THE STRICT MATHEMATICAL VIEW)

By definition the base of a logarithm must be **positive**. A negative base does not work because $\left(-\frac{1}{3}\right)^x$ is not even a real number for many x; for example, $x = \frac{1}{2}$ gives the square root of a negative number. So in strict theory, $\log_{(-\frac{1}{3})} 81$ is **not defined**.

Answer: (b) −4 (the answer the PDF and the examiners expect)

TRAP. In an MCQ with these options, mark -4. Do not mark 4 : the base is a fraction, so the power must be negative to give a number as big as 81. In a written test, viva or proof question, say that a negative base is not allowed.

Q.3 Exercise Q.13 *Relation between unknowns*

If $\log_8 p = 25$ and $\log_2 q = 5$, then

(a) $p = q^{15}$ (b) $p^2 = q^3$
(c) $p = q^5$ (d) $p^3 = q$

SOLUTION

Step 1. Remove the logs. By the definition,

$$p = 8^{25} = (2^3)^{25} = 2^{75}, \qquad q = 2^5.$$

Step 2. Write p using q. Since $75 = 5 \times 15$,

$$p = 2^{75} = (2^5)^{15} = q^{15}.$$

Second method (without exponents). Change $\log_8 p$ to base 2:

$$\log_8 p = \frac{\log_2 p}{\log_2 8} = \frac{\log_2 p}{3} = 25 \Rightarrow \log_2 p = 75.$$

Then $\log_2 p = 75 = 15 \times 5 = 15\log_2 q = \log_2 q^{15}$, so $p = q^{15}$.

Answer: (a) $p = q^{15}$

KEY IDEA. When two logs have different bases, bring both to one base (here 2). Then the relation can be read off by comparing exponents.

Part B · Nested Logarithms

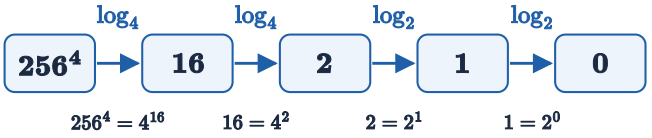
Q.4 Exercise Q.31 — *Four-level nesting*

The value of $\log_2\left[\log_2\left\{\log_4\left(\log_4 256^4\right)\right\}\right]$ is

(a) 0 (b) 1
(c) 2 (d) 4

SOLUTION

When you are **evaluating** a nested log, start from the **innermost** bracket and work outwards, one log at a time (Figure 2).



Work from the innermost bracket outwards; each arrow applies one logarithm.

Figure 2. Peeling the expression from the inside.

Layer 1. $256 = 4^4$, so $256^4 = (4^4)^4 = 4^{16}$. Therefore

$$\log_4 256^4 = \log_4 4^{16} = 16.$$

Layer 2. $\log_4 16 = \log_4 4^2 = 2.$

Layer 3. $\log_2 2 = 1.$

Layer 4. $\log_2 1 = 0$, because any base raised to the power 0 gives 1.

Answer: (a) 0

TRAP. 256^4 means $(256)^4$, not 256×4. If you ever reach the log of 0 or of a negative number partway through, you have made an arithmetic error, because such logs do not exist.

Q.5 Solved Ex. 6 = Exercise Q.28 (M.B.A. 2005, 2007)

Nested equation

If $\log_2\left[\log_3\left(\log_2 x\right)\right] = 1$, find the value of x.

(a) 0 (b) 12
(c) 128 (d) 512

SOLUTION

When you are **solving** a nested equation, do the opposite of Q.4: remove the **outermost** log first, using $\log_a N = k \iff N = a^k$.

Step 1. The outer log has base 2 and value 1, so whatever is inside equals 2^1:

$$\log_3(\log_2 x) = 2^1 = 2.$$

Step 2. The next log has base 3 and value 2:

$$\log_2 x = 3^2 = 9.$$

Step 3. The last log has base 2 and value 9:

$$x = 2^9 = 512.$$

Check. $\log_2 512 = 9$, then $\log_3 9 = 2$, then $\log_2 2 = 1$. ✓

Answer: (d) 512

TRAP. The Solved Example in the PDF stops at $\log_2 x = 9$, but the question asks for x, so take the last step. Also keep the order of the bases: the answer is $2^{(3^2)} = 2^9$, not $3^{(2^1)}$ or 2^6.

Part C · Simplifying with the Laws

Q.6 Exercise Q.39 — *Prime-factor bookkeeping*

The value of $\log 2 + 16\log\frac{16}{15} + 12\log\frac{25}{24} + 7\log\frac{81}{80}$ is

(a) 0 (b) 1
(c) 2 (d) 3

SOLUTION

Step 1. Break every fraction into primes.

$$\frac{16}{15} = \frac{2^4}{3\cdot 5}, \qquad \frac{25}{24} = \frac{5^2}{2^3\cdot 3}, \qquad \frac{81}{80} = \frac{3^4}{2^4\cdot 5}.$$

Step 2. Use the power law to expand each term. For example,

$$16\log\frac{2^4}{3\cdot 5} = 64\log 2 - 16\log 3 - 16\log 5.$$

Step 3. Collect the coefficients of $\log 2$, $\log 3$ and $\log 5$ in a table.

Term	$\log 2$	$\log 3$	$\log 5$
$\log 2$	1	0	0
$16\log\frac{16}{15}$	64	-16	-16
$12\log\frac{25}{24}$	-36	-12	24
$7\log\frac{81}{80}$	-28	28	-7
Total	1	0	1

Step 4. Put it back together.

$$\begin{aligned}\text{Value} &= 1\cdot\log 2 + 0\cdot\log 3 + 1\cdot\log 5\\ &= \log(2\times 5) = \log 10 = 1.\end{aligned}$$

Answer: (b) 1

KEY IDEA. When the fractions have big, ugly coefficients, do not multiply out huge powers. Keep a "prime ledger" of $\log 2$, $\log 3$ and $\log 5$, and most of it cancels.

Q.7 Exercise Q.36 — *Telescoping series*

The value of the following sum, taken up to 198 terms, is

$$\log_{10}\left(1+\frac{1}{2}\right) + \log_{10}\left(1+\frac{1}{3}\right) + \log_{10}\left(1+\frac{1}{4}\right) + \cdots$$

(a) 0 (b) 2
(c) 10 (d) 100

SOLUTION

Step 1. Find the general term. The terms are $1+\frac{1}{n}$ for $n = 2, 3, 4, \ldots$. Since there are 198 terms, n runs from 2 to 199. Each term can be rewritten as

$$1+\frac{1}{n} = \frac{n+1}{n}.$$

Step 2. Turn the sum of logs into the log of a product.

$$\text{Sum} = \log\left(\frac{3}{2}\times\frac{4}{3}\times\frac{5}{4}\times\cdots\times\frac{200}{199}\right).$$

Step 3. Cancel. Every numerator cancels with the next denominator (a "telescope" that folds up):

$$\frac{\cancel{3}}{2}\times\frac{\cancel{4}}{\cancel{3}}\times\frac{\cancel{5}}{\cancel{4}}\times\cdots\times\frac{200}{\cancel{199}}=\frac{200}{2}=100.$$

Step 4. $\text{Sum}=\log_{10}100=\log_{10}10^2=2.$

Answer: (b) 2

TRAP. Count the terms carefully. The first term uses $n=2$, so the 198th term uses $n=199$ and the product ends at $\frac{200}{199}$, not at $\frac{199}{198}$. Off-by-one slips here lead to the distractor options.

Q.8 Exercise Q.56 (R.R.B. 2005) *Condition with AM and GM*

$\frac{1}{2}(\log x+\log y)$ will equal $\log\left(\frac{x+y}{2}\right)$ if

(a) $y=0$ (b) $x=\sqrt{y}$
(c) $x=y$ (d) $x=\frac{y}{2}$

SOLUTION

Step 1. Simplify the left side.

$$\frac{1}{2}(\log x+\log y)=\frac{1}{2}\log(xy)=\log\sqrt{xy}.$$

Step 2. Equal logs mean equal numbers.

$$\log\sqrt{xy}=\log\frac{x+y}{2}\ \Rightarrow\ \sqrt{xy}=\frac{x+y}{2}.$$

Step 3. Square and simplify.

$$4xy=x^2+2xy+y^2$$
$$0=x^2-2xy+y^2=(x-y)^2$$

So $x-y=0$, which means $x=y$.

Answer: (c) x = y

KEY IDEA. $\sqrt{xy}$ is the geometric mean (GM) and $\frac{x+y}{2}$ is the arithmetic mean (AM). We always have AM $\geq$ GM, with equality only when the numbers are equal. Recognising this gives the answer in seconds. Option (a) is impossible, because $\log 0$ does not exist.

Q.9 Exercise Q.60 (and Q.61, M.B.A. 2007)

log x + log y = log(x + y)

If $\log x+\log y=\log(x+y)$, then

(a) $x=y$ (b) $xy=1$
(c) $y=\frac{x-1}{x}$ (d) $y=\frac{x}{x-1}$

SOLUTION

Step 1. Use the product law on the left: $\log(xy)=\log(x+y)$.

Step 2. A log never takes the same value twice (Figure 1), so the numbers inside must be equal:

$$xy=x+y.$$

Step 3. Make y the subject.

$$xy-y=x\ \Rightarrow\ y(x-1)=x\ \Rightarrow\ y=\frac{x}{x-1}.$$

Companion (Exercise Q.61). Suppose $\log\frac{a}{b}+\log\frac{b}{a}=\log(a+b)$. Then the left side is

$$\log\left(\frac{a}{b}\times\frac{b}{a}\right)=\log 1=0.$$

So $\log(a+b)=0$, which gives $a+b=10^0=1$.

Answer: (d) $y=\frac{x}{x-1}$; and for Q.61, $a+b=1$

TRAP. $\log(x+y)$ is **not** $\log x+\log y$; that is why this is a condition and not an identity. For y to be positive we need $x>1$, a fact the examiners can test separately.

Q.10 Exercise Q.59 (with Solved Ex. 2(iii)) *Identity $a^{\log_a x}=x$*

The value of $16^{\log_4 5}$ is

(a) $\frac{5}{64}$ (b) 5
(c) 16 (d) 25

SOLUTION

Why $a^{\log_a x}=x$. Let $m=\log_a x$. By the definition, $a^m=x$, so $a^{\log_a x}=x$. The power $\log_a x$ is exactly the number that turns a into x.

Step 1. Make the outer base equal to the log's base. Write $16=4^2$:

$$16^{\log_4 5}=\left(4^2\right)^{\log_4 5}=4^{2\log_4 5}.$$

Step 2. Move the 2 inside the log. $2\log_4 5=\log_4 5^2=\log_4 25$.

Step 3. Apply the identity. $4^{\log_4 25}=25$.

Companion (Solved Ex. 2(iii)). In the same way, $36^{\log_6 4}=6^{2\log_6 4}=6^{\log_6 16}=16$.

Answer: (d) 25

TRAP. $\left(4^2\right)^{\log_4 5}$ is **not** $2\times 4^{\log_4 5}=10$. The rule is $(a^m)^n=a^{mn}$: the 2 multiplies the *exponent*, and inside the log it becomes a power of 5.

Part D · Reciprocal Sums and Chain Products

Q.11 Solved Ex. 9 (M.A.T. 2005) *Reciprocal sum*

Simplify: $\frac{1}{\log_{xy}(xyz)}+\frac{1}{\log_{yz}(xyz)}+\frac{1}{\log_{zx}(xyz)}$.

SOLUTION

Step 1. Flip each fraction. The reciprocal law $\frac{1}{\log_a b}=\log_b a$ swaps the base and the number:

$$\frac{1}{\log_{xy}(xyz)}=\log_{xyz}(xy).$$

Do the same for the other two terms. Now all three logs have the **same base**, xyz:

$$\text{Sum}=\log_{xyz}(xy)+\log_{xyz}(yz)+\log_{xyz}(zx).$$

Step 2. Add them using the product law.

$$\text{Sum}=\log_{xyz}(xy\cdot yz\cdot zx)$$
$$=\log_{xyz}(x^2y^2z^2)=\log_{xyz}(xyz)^2$$
$$=2\log_{xyz}(xyz)=2\times 1=2.$$

Answer: 2

KEY IDEA. If the same expression appears *inside* every log of the denominators, flip the fractions. That expression becomes the common base, and the logs can then be added. Exercise Q.57 works the same way:

$$\frac{1}{\log_3 60}+\frac{1}{\log_4 60}+\frac{1}{\log_5 60}=\log_{60}(3\cdot 4\cdot 5)=1.$$

Q.12 Exercise Q.64 (I.I.F.T. 2005) *Hidden* $1=\log_a a$

$$\frac{1}{\log_a bc+1}+\frac{1}{\log_b ca+1}+\frac{1}{\log_c ab+1}$$ is equal to

(a) 1 (b) $\frac{3}{2}$

(c) 2 (d) 3

SOLUTION

Step 1. Write the "1" as a log. In the first denominator the base is a, so write $1=\log_a a$:

$$\log_a bc+1=\log_a bc+\log_a a=\log_a(abc).$$

In the same way, $\log_b ca+1=\log_b(abc)$ and $\log_c ab+1=\log_c(abc)$.

Step 2. Flip, as in Q.11.

$$\text{Sum}=\frac{1}{\log_a abc}+\frac{1}{\log_b abc}+\frac{1}{\log_c abc}$$
$$=\log_{abc} a+\log_{abc} b+\log_{abc} c.$$

Step 3. Add.

$$\text{Sum}=\log_{abc}(a\cdot b\cdot c)=1.$$

Answer: (a) 1

KEY IDEA. A bare "+1" next to $\log_a(\dots)$ is almost always $\log_a a$ in disguise. Absorbing it makes every term contain the same product abc. We will use this result again in Q.37.

Q.13 Exercise Q.65 *Ratio bases*

The value of $\frac{1}{\log_{\frac{p}{q}} x}+\frac{1}{\log_{\frac{q}{r}} x}+\frac{1}{\log_{\frac{r}{p}} x}$ is

(a) 0 (b) 1

(c) 2 (d) 3

SOLUTION

Step 1. Flip each term, so that x becomes the base:

$$\text{Sum}=\log_x\frac{p}{q}+\log_x\frac{q}{r}+\log_x\frac{r}{p}.$$

Step 2. Combine.

$$\text{Sum}=\log_x\left(\frac{p}{q}\times\frac{q}{r}\times\frac{r}{p}\right)=\log_x 1=0.$$

Answer: (a) 0

TRAP. Do not try to expand $\log_{\frac{p}{q}} x$ with the quotient law; that law works on the number inside the log, not on the base. Flipping first is the only clean route.

Q.14 Solved Ex. 10 (M.A.T. 2006) *Cyclic product*

If $\log_a b=\frac{1}{2}$, $\log_b c=\frac{1}{3}$ and $\log_c a=\frac{k}{5}$, find the value of k.

SOLUTION

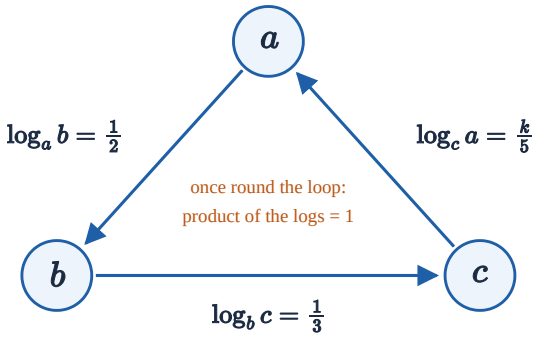


Figure 3. The bases go round a closed loop $a\to b\to c\to a$.

Step 1. Write each log with change of base.

$$\frac{\log b}{\log a}=\frac{1}{2},\qquad \frac{\log c}{\log b}=\frac{1}{3},\qquad \frac{\log a}{\log c}=\frac{k}{5}.$$

Step 2. Multiply the three equations. On the left, everything cancels:

$$\frac{\log b}{\log a}\times\frac{\log c}{\log b}\times\frac{\log a}{\log c}=1.$$

So the product of the right-hand sides is also 1:

$$\frac{1}{2}\times\frac{1}{3}\times\frac{k}{5}=1\ \Rightarrow\ \frac{k}{30}=1\ \Rightarrow\ k=30.$$

Check by the chain law. $\log_a c=\log_a b\cdot\log_b c=\frac{1}{6}$, so $\log_c a=6$. Then $\frac{k}{5}=6$ gives $k=30$. ✓

Answer: k = 30

KEY IDEA. Whenever the bases form a closed loop, such as $a\to b\to c\to a$, the product of the logs is 1. Spot the loop and the question becomes one line.

Q.15 Exercise Q.67 *Exponential loop*

If $a=b^x$, $b=c^y$ and $c=a^z$, then the value of xyz is

(a) -1 (b) 0

(c) 1 (d) abc

SOLUTION

Method 1 (logs). Turning each equation into log form gives

$$x=\log_b a,\qquad y=\log_c b,\qquad z=\log_a c.$$
$$xyz=\frac{\log a}{\log b}\times\frac{\log b}{\log c}\times\frac{\log c}{\log a}=1.$$

Method 2 (substitution, no logs at all). Start from a and keep substituting:

$$a=b^x=(c^y)^x=c^{xy}=(a^z)^{xy}=a^{xyz}.$$

So $a^1=a^{xyz}$. Since $a\neq 1$, the powers must be equal: $xyz=1$.

Answer: (c) 1

NOTE. This is Q.14 in disguise: $a\to b\to c\to a$ is again a closed loop. Its additive cousin, "if $a^x=b^y=c^z$ and $b^2=ac$, then $\frac{1}{x}+\frac{1}{z}=\frac{2}{y}$", is a favourite of Bangladeshi board and admission tests. You can prove it by taking logs in the same way.

Q.16 Exercise Q.58 — *Long chain*

The value of the following product is

$$(\log_3 4)(\log_4 5)(\log_5 6)(\log_6 7)(\log_7 8)(\log_8 9)$$

(a) 2 (b) 7
(c) 8 (d) 33

SOLUTION

Step 1. Write each factor as a fraction.

$$\frac{\log 4}{\log 3}\times\frac{\log 5}{\log 4}\times\frac{\log 6}{\log 5}\times\frac{\log 7}{\log 6}\times\frac{\log 8}{\log 7}\times\frac{\log 9}{\log 8}$$

Step 2. Cancel. Each numerator cancels the next denominator, leaving only the first denominator and the last numerator:

$$=\frac{\log 9}{\log 3}=\frac{\log 3^2}{\log 3}=\frac{2\log 3}{\log 3}=2.$$

The chain law says the same thing in one line: $\log_3 4\cdot\log_4 5\cdots\log_8 9=\log_3 9=2$.

Answer: (a) 2

KEY IDEA. In a chain, keep only the **first base** and the **last number**.

Q.17 Exercise Q.69 (M.B.A. 2008) — *Common power*

If $a=b^2=c^3=d^4$, then the value of $\log_a(abcd)$ would be

(a) $\log_a 1+\log_a 2+\log_a 3+\log_a 4$
(b) $\log_a 24$
(c) $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}$
(d) $1+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}$

SOLUTION

Step 1. Write b, c and d as powers of a.

$$b^2=a\Rightarrow b=a^{\frac{1}{2}},\qquad c^3=a\Rightarrow c=a^{\frac{1}{3}},$$
$$d^4=a\Rightarrow d=a^{\frac{1}{4}}.$$

Step 2. Multiply. When we multiply powers of the same base, the exponents add:

$$abcd=a^1\cdot a^{\frac{1}{2}}\cdot a^{\frac{1}{3}}\cdot a^{\frac{1}{4}}=a^{1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}}.$$

Step 3. Take $\log_a$.

$$\log_a(abcd)=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}=\frac{25}{12}.$$

Answer: (c) $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}$, which equals $\frac{25}{12}$

TRAP. Options (a) and (b) are equal to each other ($\log_a 1+\log_a 2+\log_a 3+\log_a 4=\log_a 24$). If two options are always equal, neither can be the single correct answer.

Part E · Expressing One Log through Others

Q.18 Exercise Q.40 (M.A.T. 2002) — *Switching base*

If $\log_a(ab)=x$, then $\log_b(ab)$ is

(a) $\frac{1}{x}$ (b) $\frac{x}{x+1}$
(c) $\frac{x}{1-x}$ (d) $\frac{x}{x-1}$

SOLUTION

Step 1. Split the given log.

$$\log_a(ab)=\log_a a+\log_a b=1+\log_a b=x,$$

so $\log_a b=x-1$.

Step 2. Flip it.

$$\log_b a=\frac{1}{\log_a b}=\frac{1}{x-1}.$$

Step 3. Split the required log the same way.

$$\log_b(ab)=\log_b a+\log_b b=\frac{1}{x-1}+1$$
$$=\frac{1+(x-1)}{x-1}=\frac{x}{x-1}.$$

Answer: (d) $\frac{x}{x-1}$

KEY IDEA. Break every log into "1 + a simple log". The only link between $\log_a b$ and $\log_b a$ is that they are reciprocals. Quick check: if $a=b$, then $x=2$, and the formula correctly gives $\frac{2}{1}=2$.

Q.19 Exercise Q.53 — *Classic "in terms of a"*

If $\log_{12}27=a$, then $\log_6 16$ is

(a) $\frac{3-a}{4(3+a)}$ (b) $\frac{3+a}{4(3-a)}$
(c) $\frac{4(3+a)}{3-a}$ (d) $\frac{4(3-a)}{3+a}$

SOLUTION

Plan. Both 12, 27 and 6, 16 are built only from the primes 2 and 3. So everything depends on one quantity, $t=\log_2 3=\frac{\log 3}{\log 2}$. Find t from the given information, then use it in the required log.

Step 1. Express a through t.

$$a=\frac{\log 27}{\log 12}=\frac{3\log 3}{2\log 2+\log 3}.$$

Divide the top and the bottom by $\log 2$:

$$a=\frac{3t}{2+t}.$$

Step 2. Solve for t.

$$a(2+t)=3t$$
$$2a+at=3t$$
$$2a=t(3-a)\ \Rightarrow\ t=\frac{2a}{3-a}.$$

Step 3. Express the required log through t.

$$\log_6 16=\frac{\log 16}{\log 6}=\frac{4\log 2}{\log 2+\log 3}=\frac{4}{1+t}.$$

Step 4. Substitute.

$$1+t=1+\frac{2a}{3-a}=\frac{3-a+2a}{3-a}=\frac{3+a}{3-a}$$
$$\therefore\ \log_6 16=4\times\frac{3-a}{3+a}=\frac{4(3-a)}{3+a}.$$

Numerical check. A calculator gives $a=\log_{12}27\approx 1.3263$, and then

$$\frac{4(3-1.3263)}{3+1.3263} \approx 1.5474 \approx \log_6 16. \checkmark$$

Answer: (d) $\dfrac{4(3-a)}{3+a}$

KEY IDEA. When all the numbers use only two primes, reduce everything to the single ratio $\log_2 3$. This one idea solves a whole family of JEE, CAT and admission-test questions.

Q.20 Exercise Q.42 (M.B.A. 2010) *Using log 5 = 1 − log 2*

If $\log_{10} 2 = a$ and $\log_{10} 3 = b$, then $\log_5 12$ equals

(a) $\dfrac{a+b}{1+a}$ (b) $\dfrac{2a+b}{1+a}$

(c) $\dfrac{a+2b}{1+a}$ (d) $\dfrac{2a+b}{1-a}$

SOLUTION

Step 1. Change to base 10, because the given values are in base 10:

$$\log_5 12 = \frac{\log 12}{\log 5}.$$

Step 2. Numerator. Since $12 = 2^2 \times 3$,

$$\log 12 = 2\log 2 + \log 3 = 2a + b.$$

Step 3. Denominator. 5 is not built from 2 and 3, but 5 is 10 divided by 2:

$$\log 5 = \log 10 - \log 2 = 1 - a.$$

Step 4. $\log_5 12 = \dfrac{2a+b}{1-a}$.

Answer: (d) $\dfrac{2a+b}{1-a}$

TRAP. Never write $\log 5 = \log 2 + \log 3$; that would be $\log 6$. Always use $\log 5 = 1 - \log 2$. The wrong denominators $1 + a$ in options (a) to (c) come from exactly this sign slip.

Q.21 Exercise Q.80 (M.A.T. 2002) *Numerical change of base*

If $\log 2 = 0.3010$ and $\log 3 = 0.4771$, the value of $\log_5 512$ is

(a) 2.870 (b) 2.967

(c) 3.876 (d) 3.912

SOLUTION

Step 1. Change to base 10. $512 = 2^9$, so

$$\log_5 512 = \frac{\log 2^9}{\log 5} = \frac{9\log 2}{1 - \log 2}.$$

Step 2. Substitute.

$$\frac{9 \times 0.3010}{1 - 0.3010} = \frac{2.709}{0.699} \approx 3.876.$$

Sanity check before dividing. $5^3 = 125$ and $5^4 = 625$. Since 512 lies between them and much nearer to 625, the answer must be between 3 and 4 and close to 4. That rules out (a) and (b) straight away.

Answer: (c) 3.876

TRAP. The value of $\log 3$ is a **distractor**; it is not needed. Examiners often give extra data to make you waste time looking for a use for it.

Part F · Logarithmic Equations

Q.22 Solved Ex. 7 (R.R.B. 2006) *Quadratic inside a log*

If $\log_{10}(x^2 - 6x + 45) = 2$, find the value of x.

SOLUTION

Step 1. Remove the log. $\log_{10} N = 2$ means $N = 10^2$:

$$x^2 - 6x + 45 = 100 \Rightarrow x^2 - 6x - 55 = 0.$$

Step 2. Factorise. We need two numbers whose product is -55 and whose sum is -6; these are -11 and 5:

$$x^2 - 11x + 5x - 55 = 0 \Rightarrow (x-11)(x+5) = 0.$$

So $x = 11$ or $x = -5$.

Step 3. Domain check (never skip this). The log needs $x^2 - 6x + 45 > 0$. Completing the square, $x^2 - 6x + 45 = (x-3)^2 + 36$, which is positive for **every** real x. So both roots are acceptable. Indeed, both make the expression equal to exactly 100.

Answer: x = 11 or x = −5

TRAP. Do not throw away $x = -5$ just because it is negative. The condition is that the **number inside the log** is positive, not that x is positive. Compare this with Q.23, where a root really does have to be rejected.

Q.23 Exercise Q.55 (M.B.A. 2007) *Domain decides*

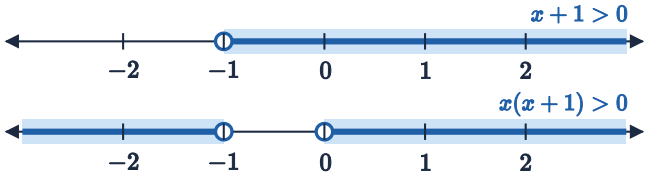
If $\log_5(x^2 + x) - \log_5(x+1) = 2$, then the value of x is

(a) 5 (b) 10

(c) 25 (d) 32

SOLUTION

Step 1. Domain first. Both logs must exist: $x + 1 > 0$ and $x^2 + x = x(x+1) > 0$. Figure 4 shows that the only x satisfying both is $x > 0$.



Common part (both conditions true): only x > 0 survives.

Figure 4. Finding the domain on number lines.

Step 2. Use the quotient law.

$$\log_5 \frac{x^2+x}{x+1} = 2 \Rightarrow \log_5 \frac{x(x+1)}{x+1} = 2.$$

Because $x + 1 \neq 0$ in the domain, we may cancel it: $\log_5 x = 2$.

Step 3. $x = 5^2 = 25$. Since $25 > 0$, it lies in the domain. ✓

Answer: (c) 25

TRAP. $\log A - \log B = \log \dfrac{A}{B}$, **not** $\log(A - B)$. Treating it as $\log_5 [(x^2 + x) - (x + 1)]$ gives $x^2 - 1 = 25$, a wrong equation with no option to match.

Q.24 Exercise Q.70 (M.B.A. 2010) *Mixed bases*

If $\log_3 x + \log_9 x^2 + \log_{27} x^3 = 9$, then x equals

(a) 3 (b) 9
(c) 27 (d) None of these

SOLUTION

Step 1. Bring every term to base 3. Use $\log_{a^q} x^p = \frac{p}{q}\log_a x$. This holds because $\log_{a^q} x^p = \frac{p \log x}{q \log a}$.

$$\log_9 x^2 = \log_{3^2} x^2 = \frac{2}{2}\log_3 x = \log_3 x$$

$$\log_{27} x^3 = \log_{3^3} x^3 = \frac{3}{3}\log_3 x = \log_3 x$$

Step 2. Add.

$$3\log_3 x = 9 \;\Rightarrow\; \log_3 x = 3 \;\Rightarrow\; x = 3^3 = 27.$$

Answer: (c) 27

KEY IDEA. When the bases are powers of one number (3, 9, 27 or 2, 4, 8, 16), reduce everything to the smallest base. Exercise Q.44 is the same pattern: $\log_4 x + \log_2 x = 6$ gives $\frac{1}{2}\log_2 x + \log_2 x = 6$, so $\log_2 x = 4$ and $x = 16$.

Q.25 Exercise Q.71 (M.B.A. 2009) *Surd equation inside a log*

If $\log_7 \log_5 \left(\sqrt{x+5} + \sqrt{x}\right) = 0$, what is the value of x?

(a) 2 (b) 3
(c) 4 (d) 5

SOLUTION

Step 1. Peel the outer log. $\log_7(\ldots) = 0$ means the inside equals $7^0 = 1$:

$$\log_5 \left(\sqrt{x+5} + \sqrt{x}\right) = 1 \;\Rightarrow\; \sqrt{x+5} + \sqrt{x} = 5. \quad \text{(i)}$$

Step 2. A neat trick instead of squaring. Multiply the conjugates:

$$\left(\sqrt{x+5} + \sqrt{x}\right)\left(\sqrt{x+5} - \sqrt{x}\right) = (x+5) - x = 5.$$

Since the first bracket equals 5, the second bracket must equal 1:

$$\sqrt{x+5} - \sqrt{x} = 1. \quad \text{(ii)}$$

Step 3. Add (i) and (ii).

$$2\sqrt{x+5} = 6 \;\Rightarrow\; \sqrt{x+5} = 3 \;\Rightarrow\; x + 5 = 9 \;\Rightarrow\; x = 4.$$

Check. Working outwards from $x = 4$:

$$\sqrt{9} + \sqrt{4} = 5, \quad \log_5 5 = 1, \quad \log_7 1 = 0 \;\checkmark$$

Answer: (c) 4

NOTE. The PDF squares twice, which takes about six lines. The conjugate trick needs only two. Remember that $(\sqrt{A} + \sqrt{B})(\sqrt{A} - \sqrt{B}) = A - B$.

Q.26 Exercise Q.91 (I.I.F.T. 2012) *Logs in AP*

If $\log 3$, $\log(3^x - 2)$ and $\log(3^x + 4)$ are in arithmetic progression, then x is equal to

(a) $\frac{8}{3}$ (b) $\log_3 8$
(c) $\log_2 3$ (d) 8

SOLUTION

Step 1. AP condition. For three terms A, B, C in AP, $2B = A + C$:

$$2\log(3^x - 2) = \log 3 + \log(3^x + 4).$$

Step 2. Remove the logs. Using the power and product laws,

$$\log(3^x - 2)^2 = \log\left[3(3^x + 4)\right]$$

$$\Rightarrow\; (3^x - 2)^2 = 3(3^x + 4).$$

Step 3. Substitute $t = 3^x$. Note that $t > 0$, and the log $\log(3^x - 2)$ needs $t > 2$.

$$t^2 - 4t + 4 = 3t + 12$$

$$t^2 - 7t - 8 = 0$$

$$(t - 8)(t + 1) = 0$$

So $t = 8$ or $t = -1$. Since $t = 3^x$ can never be negative, we reject $t = -1$.

Step 4. $3^x = 8 \;\Rightarrow\; x = \log_3 8 \; (= 3\log_3 2 \approx 1.89)$.

Check. If $3^x = 8$, the three terms are $\log 3$, $\log 6$ and $\log 12$, i.e. $\log 3$, $\log 3 + \log 2$, $\log 3 + 2\log 2$. They are in AP with common difference $\log 2$. ✓

Answer: (b) $\log_3 8$

CAUTION: ERROR IN THE PDF. The printed key marks (c), and the PDF's own working is garbled. The correct value is $x = \log_3 8$; the check above confirms it. Logs in AP mean the numbers themselves are in **GP**: 3, 6, 12. This is a quick way to spot the answer.

Part G · Characteristic, Mantissa and Digits

Q.27 Exercise Q.85 (M.B.A. 2009) *Negative log to bar form*

If the logarithm of a number is -3.153, what are its characteristic and mantissa?

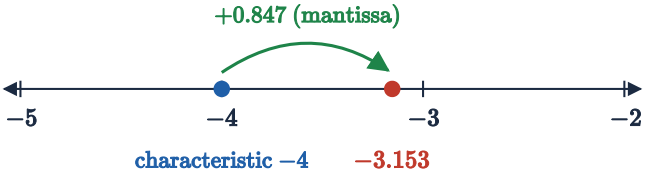
(a) -4, 0.847 (b) -3, -0.153
(c) 4, -0.847 (d) 3, -0.153

SOLUTION

Rule. The mantissa must lie in $[0, 1)$, so it can **never be negative**. This immediately rules out (b), (c) and (d).

Step 1. Step down to the next lower integer. The integer just below -3.153 is -4 (Figure 5).

$$-3.153 = -4 + (4 - 3.153) = -4 + 0.847.$$



Step DOWN to the integer on the left, then climb back up.

Figure 5. Writing -3.153 as an integer plus a positive decimal.

Step 2. So the characteristic is -4 and the mantissa is 0.847. In bar notation, $-3.153 = \bar{4}.847$.

Answer: (a) Characteristic = −4, mantissa = 0.847

> **TRAP.** "Characteristic -3, mantissa -0.153" only copies the digits and breaks the rule that the mantissa is positive. For a negative log, the characteristic is always **one more in size** than the integer you see: -3.153 goes to -4.

Q.28 Exercise Q.84 — *Bar arithmetic*

If $\log(0.57) = \bar{1}.756$, then the value of the following is

$$\log 57 + \log(0.57)^3 + \log\sqrt{0.57}$$

(a) 0.902 (b) $\bar{2}.146$
(c) 1.902 (d) $\bar{1}.146$

SOLUTION

Step 1. Decode the bar. $\bar{1}.756$ means $-1 + 0.756 = -0.244$. So $\log 0.57 = -0.244$.

Step 2. Find $\log 57$. Since $57 = 0.57 \times 100$,

$$\log 57 = \log 0.57 + \log 100 = -0.244 + 2 = 1.756.$$

The digits 57 are the same, so the mantissa .756 is the same; only the characteristic changes.

Step 3. The other two terms.

$$\log(0.57)^3 = 3 \times (-0.244) = -0.732$$

$$\log\sqrt{0.57} = \frac{1}{2} \times (-0.244) = -0.122$$

Step 4. Add.

$$1.756 - 0.732 - 0.122 = 0.902.$$

Method 2 (staying in bar form). Tripling:

$$3 \times \bar{1}.756 = \bar{3} + 2.268 = \bar{1}.268.$$

To halve, first make the integer part even, then halve both parts:

$$\bar{1}.756 = \bar{2} + 1.756 \;\Rightarrow\; \tfrac{1}{2}\left(\bar{1}.756\right) = \bar{1}.878.$$

Adding the three logs:

$$1.756 + \bar{1}.268 + \bar{1}.878$$
$$= (1 - 1 - 1) + 1.902 = 0.902.$$

Answer: (a) 0.902

> **KEY IDEA.** Converting a bar log to an ordinary negative decimal, as in $\bar{1}.756 = -0.244$, makes the arithmetic safe. Convert back to bar form only if the answer options require it.

Q.29 Exercise Q.89 (also Solved Ex. 13) — *Number of digits*

If $\log 2 = 0.30103$ and $\log 3 = 0.47712$, then the number of digits in 6^{20} is

(a) 15 (b) 16
(c) 17 (d) 18

SOLUTION

Step 1. Take the log.

$$\log 6^{20} = 20 \log 6 = 20(\log 2 + \log 3)$$
$$= 20 \times 0.77815 = 15.563.$$

Step 2. Read the characteristic. The characteristic is 15, so the number of digits is $15 + 1 = 16$.

Why "+1"? $\log 6^{20} = 15.563$ means $6^{20} = 10^{15.563}$, which lies between 10^{15} (the smallest 16-digit number) and 10^{16} (the smallest 17-digit number). See Figure 6.

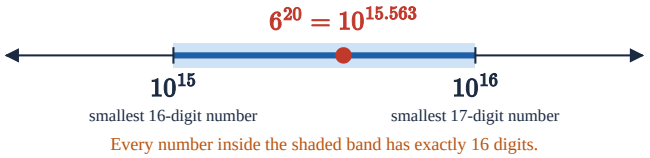


Figure 6. A characteristic of 15 puts the number in the 16-digit band.

In the same way, Solved Ex. 13 gives $\log 2^{56} = 16.857$, so 2^{56} has 17 digits.

Answer: (b) 16

> **TRAP.** Option (a), 15, is the characteristic, not the number of digits. Always add 1 to the characteristic.

Q.30 Exercise Q.90 (A.A.O. 2009) — *Digits without tables*

The number of digits in $4^9 \times 5^{17}$, when expressed in usual form, is

(a) 16 (b) 17
(c) 18 (d) 19

SOLUTION

Step 1. Make tens. Each 2 paired with a 5 makes a 10. First, $4^9 = (2^2)^9 = 2^{18}$, so

$$4^9 \times 5^{17} = 2^{18} \times 5^{17} = 2 \times (2^{17} \times 5^{17}) = 2 \times 10^{17}.$$

Step 2. Count. 2×10^{17} is the digit 2 followed by 17 zeros, so it has $1 + 17 = 18$ digits.

Log check.

$$\log(2 \times 10^{17}) = 17 + \log 2 = 17.3010.$$

The characteristic is 17, so there are 18 digits. ✓

Answer: (c) 18

> **KEY IDEA.** When the powers of 2 and 5 are close to each other, pair them into 10s first. The answer then needs no log table at all.

Part H · Extra Questions (not in the PDF)

These eight questions cover patterns that the PDF does not test but that appear regularly in BUET, DU, Engineering and Medical admission tests, JEE (Main), CAT, and the HSC and SSC higher-math boards. They round off the chapter.

Q.31 Extra — *Variable in base and exponent*

Solve: $x^{\log x} = 100x$.

SOLUTION

Step 1. Take $\log_{10}$ of both sides. We need $x > 0$. Using the power law on the left and the product law on the right,

$$\log x \cdot \log x = \log 100 + \log x.$$

Step 2. Substitute $t = \log x$.

$$t^2 = 2 + t \;\Rightarrow\; t^2 - t - 2 = 0 \;\Rightarrow\; (t-2)(t+1) = 0.$$

Step 3. Convert back.

$$t = 2 \Rightarrow x = 10^2 = 100; \qquad t = -1 \Rightarrow x = 10^{-1} = \frac{1}{10}.$$

Check.

$$x = 100: \quad 100^2 = 10000 = 100 \times 100 \;\checkmark$$
$$x = \frac{1}{10}: \quad \left(\frac{1}{10}\right)^{-1} = 10 = 100 \times \frac{1}{10} \;\checkmark$$

Answer: x = 100 or $x = \frac{1}{10}$

KEY IDEA. When x appears in both the base and the exponent, take logs. The equation becomes a quadratic in $\log x$.

Q.32 Extra — *Swap identity*

Solve: $3^{\log x} + x^{\log 3} = 54$.

SOLUTION

Step 1. Show that the two terms are equal. Take the log of each:

$$\log\left(x^{\log 3}\right) = \log 3 \cdot \log x, \qquad \log\left(3^{\log x}\right) = \log x \cdot \log 3.$$

The logs are equal, so the numbers are equal: $x^{\log 3} = 3^{\log x}$. This is the swap law $x^{\log_a y} = y^{\log_a x}$.

Step 2. Solve.

$$2 \cdot 3^{\log x} = 54 \;\Rightarrow\; 3^{\log x} = 27 = 3^3 \;\Rightarrow\; \log x = 3.$$

Step 3. $x = 10^3 = 1000$.

Check.

$$27 + 1000^{\log 3} = 27 + 10^{3\log 3} = 27 + 3^3 = 54 \;\checkmark$$

Answer: x = 1000

NOTE. The same identity explains Exercise-style questions such as "$7^{\log 3} - 3^{\log 7} = ?$". The answer is 0, because the two terms are equal.

Q.33 Extra — *Unknown in the base*

Solve: $\log_x 2 \cdot \log_{2x} 2 = \log_{4x} 2$.

SOLUTION

Step 1. Domain. Each base must be positive and different from 1: $x > 0$, and $x \neq 1$, $2x \neq 1$, $4x \neq 1$ (Figure 7).



Allowed: x > 0, except the hollow points.

Figure 7. Domain for an unknown base.

Step 2. Flip every log to base 2. Let $t = \log_2 x$. Then

$$\log_x 2 = \frac{1}{\log_2 x} = \frac{1}{t},$$
$$\log_{2x} 2 = \frac{1}{\log_2 2x} = \frac{1}{1+t},$$
$$\log_{4x} 2 = \frac{1}{\log_2 4x} = \frac{1}{2+t}.$$

Step 3. Solve.

$$\frac{1}{t(1+t)} = \frac{1}{2+t} \;\Rightarrow\; t + t^2 = 2 + t \;\Rightarrow\; t^2 = 2.$$

So $t = \pm\sqrt{2}$. Neither value is 0, -1 or -2, so the domain is respected.

Step 4. $x = 2^{\sqrt{2}}$ or $x = 2^{-\sqrt{2}}$.

Answer: $x = 2^{\sqrt{2}}$ or $x = 2^{-\sqrt{2}}$

KEY IDEA. When the unknown is in the base, flip every log so that the fixed number becomes the base. Then put $t = \log(\text{unknown})$.

Q.34 Extra — *Log inequality, base below 1*

Solve: $\log_{\frac{1}{2}}(2x - 1) > -2$.

SOLUTION

Step 1. Domain. $2x - 1 > 0 \Rightarrow x > \frac{1}{2}$.

Step 2. Write -2 as a log with the same base.

$$-2 = \log_{\frac{1}{2}}\left(\frac{1}{2}\right)^{-2} = \log_{\frac{1}{2}} 4.$$

Step 3. Remove the log and flip the sign. Base $\frac{1}{2}$ is less than 1, so the function is **decreasing**: a bigger log comes from a smaller number. Put $u = 2x - 1$ (Figure 8):

$$\log_{\frac{1}{2}} u > \log_{\frac{1}{2}} 4 \;\Longrightarrow\; u < 4.$$

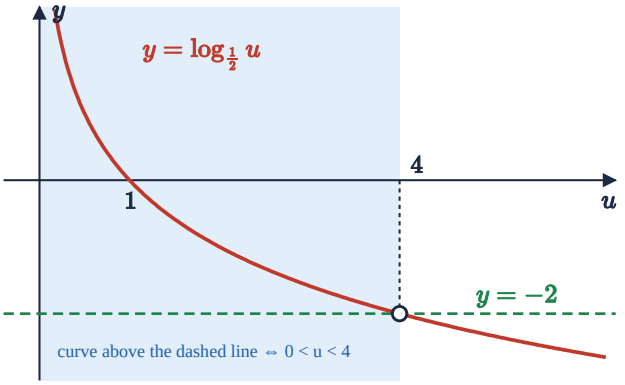


Figure 8. A falling curve reverses the inequality.

Step 4. Combine with the domain. $0 < 2x - 1 < 4$, so $1 < 2x < 5$ and

$$\frac{1}{2} < x < \frac{5}{2}.$$

Answer: $\frac{1}{2} < x < \frac{5}{2}$

TRAP. If you keep the sign, you get $x > \frac{5}{2}$, which is exactly the wrong half. Compare with $\log_2(2x - 1) > 2$: there the base is greater than 1, so the sign stays and $x > \frac{5}{2}$.

Q.35 Extra — *Comparing logs without tables*

Which is greater: $\log_2 3$ or $\log_3 5$?

SOLUTION

Idea. Find a simple number that lies between the two logs. Try $\frac{3}{2}$.

Step 1. Compare $\log_2 3$ with $\frac{3}{2}$.

$$\log_2 3 > \frac{3}{2} \iff 3 > 2^{\frac{3}{2}} \iff 3^2 > 2^3 \iff 9 > 8.$$

The last statement is true, so $\log_2 3 > \frac{3}{2}$.

Step 2. Compare $\log_3 5$ with $\frac{3}{2}$.

$$\log_3 5 < \frac{3}{2} \iff 5 < 3^{\frac{3}{2}} \iff 5^2 < 3^3 \iff 25 < 27.$$

This is also true, so $\log_3 5 < \frac{3}{2}$.

Step 3. $\log_3 5 < \frac{3}{2} < \log_2 3$.

Answer: $\log_2 3$ is greater

KEY IDEA. Squaring (or cubing) both sides removes fractional powers, so a log comparison becomes a comparison of whole numbers. For a check: $\log_2 3 \approx 1.585$ and $\log_3 5 \approx 1.465$.

Q.36 Extra — *Zeros after the decimal point*

Given $\log 2 = 0.30103$, how many zeros are there between the decimal point and the first significant digit of the number below? What is that first digit?

$$\left(\frac{1}{2}\right)^{50}$$

SOLUTION

Step 1. Take the log.

$$\log\left(\frac{1}{2}\right)^{50} = -50\log 2 = -15.0515.$$

Step 2. Bar form (as in Q.27):

$$-15.0515 = -16 + 0.9485 = \overline{16}.9485.$$

Step 3. Count the zeros. A characteristic of $\overline{n}$ means $n-1$ zeros after the decimal point, so there are $16 - 1 = 15$ zeros.

Step 4. First digit from the mantissa. $\log 8 = 0.9031$ and $\log 9 = 0.9542$. Since $0.9031 < 0.9485 < 0.9542$, the first significant digit is 8. In fact,

$$\left(\frac{1}{2}\right)^{50} \approx 0.000000000000000888.$$

Answer: 15 zeros; the first significant digit is 8

TRAP. Do not answer 16. The characteristic $\overline{16}$ counts one more than the number of zeros. Similarly, 0.05 has characteristic $\overline{2}$ but only one zero after the point.

Q.37 Extra — *Proof-type identity*

Let

$$x = \log_a bc, \qquad y = \log_b ca, \qquad z = \log_c ab.$$

Prove that

$$xyz = x + y + z + 2.$$

SOLUTION

Step 1. Use Q.12. As shown there, $1 + x = \log_a abc$, so

$$\frac{1}{1+x} = \log_{abc} a.$$

The same holds for y and z. Adding the three results,

$$\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = \log_{abc}(abc) = 1.$$

Step 2. Clear the denominators. Multiply both sides by the product $(1+x)(1+y)(1+z)$:

$$(1+y)(1+z) + (1+x)(1+z) + (1+x)(1+y) = (1+x)(1+y)(1+z).$$

Step 3. Expand both sides.

$$\text{LHS} = 3 + 2(x+y+z) + (xy+yz+zx)$$
$$\text{RHS} = 1 + (x+y+z) + (xy+yz+zx) + xyz$$

Step 4. Cancel the common terms. $xy+yz+zx$ cancels from both sides, leaving

$$3 + 2(x+y+z) = 1 + (x+y+z) + xyz,$$
$$\therefore\ xyz = x + y + z + 2. \blacksquare$$

KEY IDEA. Proof questions are often an MCQ trick (Q.12) followed by algebra. Spotting the link with an MCQ you already know saves the whole proof.

Q.38 Extra — *Algebraic condition to log identity*

If $a^2 + b^2 = 7ab$, where a and b are positive, prove that

$$\log\frac{a+b}{3} = \frac{1}{2}(\log a + \log b).$$

SOLUTION

Step 1. Build a perfect square. Add $2ab$ to both sides of the condition:

$$a^2 + 2ab + b^2 = 9ab \ \Rightarrow\ (a+b)^2 = 9ab.$$

Step 2. Make the left side match the target. Divide by 9:

$$\left(\frac{a+b}{3}\right)^2 = ab.$$

Step 3. Take logs. Both sides are positive, so

$$2\log\frac{a+b}{3} = \log a + \log b,$$
$$\therefore\ \log\frac{a+b}{3} = \frac{1}{2}(\log a + \log b). \blacksquare$$

KEY IDEA. Start from the target. The "3" in $\frac{a+b}{3}$ tells you to aim for $(a+b)^2 = 9ab$, and $9ab = 7ab + 2ab$. Also try the variants $a^2 + b^2 = 23ab$, which gives $\frac{a+b}{5}$, and $a^2 + b^2 = 14ab$, which gives $\frac{a+b}{4}$.